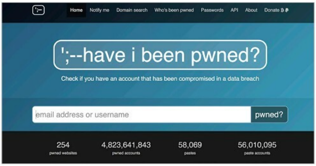

Figure 1: Hacked credentials

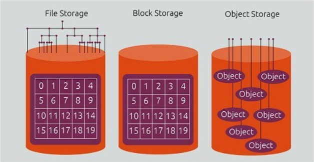

Figure 2: Object storage, file systems and block storage [Source: ubuntu.com]

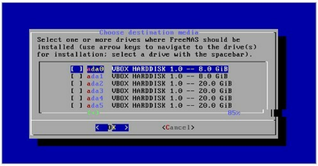

Figure 3: Selecting the boot partition [Source: freenas.org]

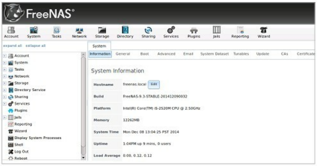

Figure 4: FreeNAS GUI [Source: freenas.org]

Block storage

Data is stored as a sequence of bytes, termed a physical record. This so called 'block' of data comprises a whole number of records. The process of putting data into blocks is termed as blocking, while the reverse is called deblocking. Blocking is widely employed when storing data to certain types of magnetic tape, Flash memory and rotating media.

File systems

These data storage structures follow a hierarchy, which controls how data is stored and retrieved. In the absence of a file system, information would simply be a large body of data with no way to isolate individual pieces of information from the whole. A file system encapsulates the complete set of rules and logic used to manage sets of data. File systems can be used on a variety of storage media, most commonly, hard disk drives (HDDs), magnetic tapes and optical discs.

Building open source storage

Software

Network Attached Storage (NAS) provides a stable and widely employed alternative for data storage and sharing across a network. It provides centralised repository of data that can be accessed by different members within the organisation. Variations include providing complete software and hardware packages serving as out-of-the-box alternatives. These include software and file systems such as Gluster, Ceph, NAS4Free, FreeNAS, and others. As an example, we will look into the general steps involved in deploying such a system by taking the case of a popular representative of the set.

FreeNAS

With enterprise-grade features, richly supported plugins, and an enterprise-ready ZFS file system, it is easy to see why FreeNAS is one of the most popular operating systems in the market for data storage.

Let's take a deeper look at file systems since they are widely used in setting up storage networks today. Building your own data storage using FreeNAS involves following a few of the following simple steps:

1. You will need to download the disk image suitable for your architecture and burn it onto either a USB stick or a CD-ROM, as per your preference.
2. Since you will be booting your new disk or machine with FreeNAS, you will need to open the BIOS settings on booting it, and set the boot preference to USB so that your system first tries to boot from the USB and, if not found, then from other attached media.
3. Once you have created the storage media with the required software, you can boot up your system and install FreeNAS in the designated partition.
4. Having set the root password, when you boot into it after installation, you will have the option of using the Web GUI to log into the system. For some users, it might be much more intuitive to use this option as compared to the console-based login.
5. Using the GUI or console, you can configure and manage